

Assignment 1: Grid World Navigation Using Q-Learning

Problem Statement

Design and implement a Q-Learning agent that learns to navigate from a start position to a goal position in a grid environment containing obstacles. The agent must discover the optimal path that reaches the goal while avoiding collisions and minimizing the number of steps.

Description

A grid world consists of a Start position (S), Goal position (G), Obstacles (X), and empty cells. The objective is to learn an optimal navigation policy that reaches the goal using the shortest safe path.

Example Environment

```
S . . . X
. . X . .
X . . . X
. . . X .
X . . . .
. . X . G
```

User Inputs

1. Number of rows (N)
2. Number of columns (M)
3. Start position, Goal position
4. Number of obstacles
5. Obstacle coordinates
6. Learning rate (α)
7. Discount factor (γ)
8. Exploration rate (ϵ)
9. Number of training episodes

Constraints

- Agent cannot move outside the grid.
- Agent cannot move through obstacles.
- Episode terminates when goal is reached or maximum steps are exceeded.
- Maximum Steps = $2 \times N \times M$

Action Space :Up, Down, Left, Right

Reward Function

- Reach Goal: +100
- Valid Move: -1
- Hit Obstacle: -100
- Invalid Move: -10

What You Have to Find

1. Optimal policy learned by Q-Learning
2. Best path from Start to Goal
3. Total reward obtained
4. Number of steps required

What You Have to Show(Display)

1. Final Q-table
2. Learned optimal path
3. Total reward per episode graph
4. Steps per episode graph
5. Comparison of early vs final performance